

Data Analytics Sample

Mapping the Emergence of AI in Genomics and Biotechnology: A Bibliometric Analysis,
2010 to 2025

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Abstract

Prepared for the Center for Security and Emerging Technology (CSET), Walsh School of Foreign Service (SFS), Georgetown University. This sample builds an end to end bibliometric pipeline on open data (OpenAlex), resolves messy affiliation records, detects emerging research clusters from paper text, uses an LLM (Claude) to annotate a random sample and measure how clean the corpus is, and translates the results into findings for a policy audience.

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Key findings

i Question. How fast is artificial intelligence spreading through genomics and biotechnology research, which research fronts are emerging fastest, and which countries are producing that work?

- **AI is moving from niche to mainstream in biotechnology.** AI-related genomics and biotechnology papers rose from 445 in 2010 to 18,221 in 2025, about 28% a year, roughly three times the growth rate of biotechnology research overall (9%). AI work went from 1.2% to 12.9% of the field, and output grew 55% in 2025 alone.
- **The United States leads on influence; China is closing the gap on volume.** Between 2010 and 2020, China’s share of world output rose from 11% to 17%, passing the EU27 (16%). The United States held about a third of output (34% in 2020) but produced 46% of the top 1 percent most cited papers.
- **Research fronts shift quickly.** Of 30 clusters detected from paper text, the fastest growing between 2013 to 2016 and 2018 to 2020 were pandemic genomics (COVID-19), single-cell analysis, AI in clinical care, and tumor immunology. Older approaches such as microarray-based gene classification lost ground.
- **US and China research remains intertwined.** About 10% of US papers in 2020 had a co-author based in China, up from 7% in 2010. This is relevant to current research security debates.
- **Data quality steps change the numbers.** Claude’s annotation of a random sample suggests about 17% of keyword matches are not really AI biology research, and an affiliation matcher, checked against a hidden holdout, recovered institutions for about a quarter of the author affiliations OpenAlex left unlinked (Section 4).

! Data coverage. Yearly totals cover 2010 to 2025. Full paper records (authors, institutions, abstracts, citations) were downloaded for 2010 to 2020, 15,502 papers, before OpenAlex’s free daily limit for requests without an API key was reached. Country, cluster, and collaboration results therefore describe 2010 to 2020. Rerunning the same pipeline with a free API key extends every figure to 2025 with no code changes.

1 Why this matters

AI tools such as protein structure predictors, genomic language models, and generative molecule designers are changing what a small team can do in biology. That capability has large benefits for medicine and agriculture, and it also raises security questions: lower barriers to engineering biological systems, concentration of capability in a few countries or companies, and research security concerns around international collaboration. Tracking where this work is growing, and who is doing it, is a first step toward informed policy.

This sample asks four questions:

1. How quickly is AI being adopted in genomics and biotechnology research?
2. Which research fronts within that space are emerging fastest?
3. How is output distributed across countries, and how is collaboration changing?
4. How reliable are the underlying data, and how much do data quality steps change the answers?

2 Data and methods

2.1 Data source

Publication records come from **OpenAlex**, a fully open index of more than 250 million scholarly works with linked authors, institutions, countries, topics, and citations. The corpus is every article, preprint, or review from 2010 to 2025 whose title or abstract contains at least one AI term *and* at least one genomics or biotechnology term (full term lists in `config.yaml`). A broader biotechnology query serves as the denominator for adoption rates.

Table 1: Corpus summary

Item	Value
Records matching the query in OpenAlex	66,339
Years with yearly totals	2010 to 2025
Full records downloaded and analyzed	15,502
Years with full records	2010 to 2020
Retrieved	2026-09-24

For 2010 to 2020 every matching record was downloaded. The pipeline also supports seeded random sampling per year with sampling weights, for corpora too large to download in full; the weights here are close to 1.

2.2 Pipeline

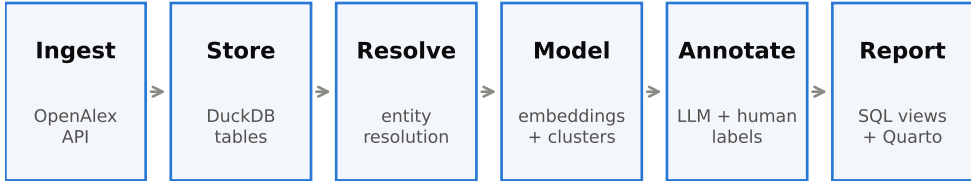


Figure 1: Data pipeline. Each box is a Python module; the same steps are orchestrated as an Airflow DAG (`days/cset_sample_dag.py`).

Counting. Country credit uses *fractional counting*: each paper is worth one unit, split equally among its authors, and an author with several country affiliations splits their share evenly. This avoids double counting large international collaborations.

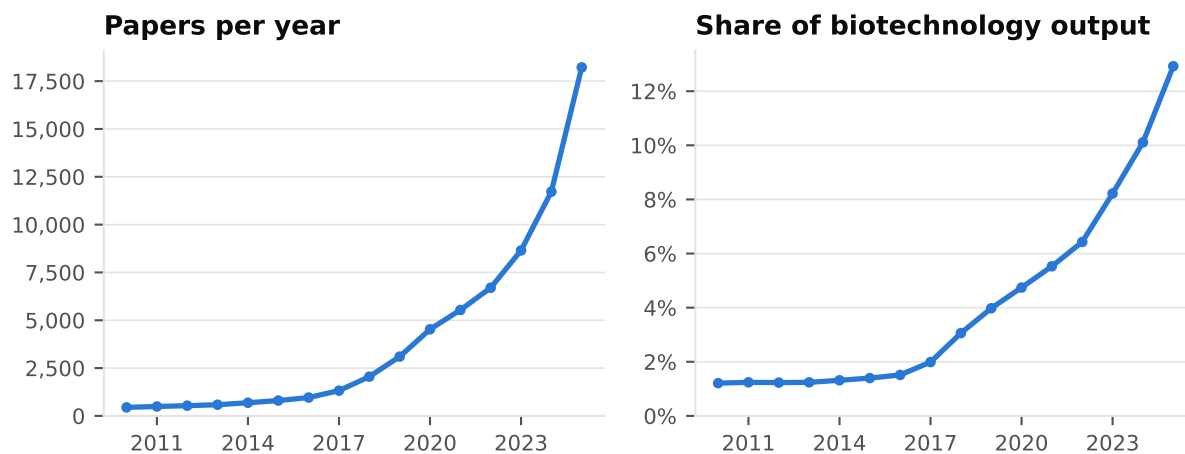
Impact. Highly cited work is measured with OpenAlex’s field and year normalized citation percentile (top 10 percent flag), so older papers and high citation fields do not dominate. The most recent year is excluded from impact measures.

Emerging topics. Titles and abstracts are represented with TF-IDF and reduced to 256 dimensions with truncated SVD (latent semantic analysis), grouped with k-means into 30 clusters, and labeled with class-based TF-IDF terms plus the papers nearest each cluster center. A cluster is flagged as emerging when its share of the corpus grew in the top quartile between the two comparison windows

(2013 to 2016 versus 2018 to 2020) and it has at least 100 papers. Each cluster was then named by reading its terms and example papers.

3 Findings

3.1 Adoption is accelerating

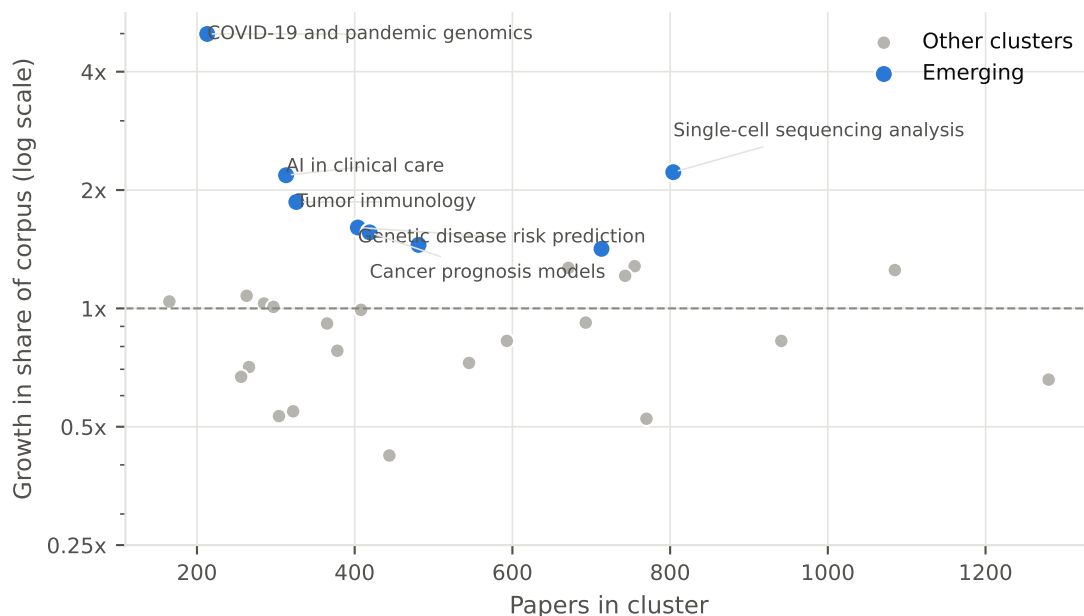


Source: OpenAlex; analysis by the author.

Figure 2: AI in genomics and biotechnology: annual papers (left) and share of all biotechnology papers (right). Two panels are used instead of a dual axis so each measure keeps its own honest scale.

Growth was steady through the early 2010s and then bent upward from about 2017, as deep learning frameworks matured and large public sequencing datasets became standard training data. The share of all biotechnology papers that use AI climbed every year in the series, which means AI is spreading faster than the field is growing, not simply riding its expansion. The largest single-year jumps came in 2025 (+55%), 2018 (+55%), 2019 (+51%). The 2025 jump follows the spread of protein structure prediction tools and genomic language models, although yearly totals alone cannot show which methods drove it.

3.2 Emerging research fronts



Source: OpenAlex; analysis by the author.

Figure 3: Research clusters by size and growth. Each dot is a cluster of papers grouped by text similarity. Clusters flagged as emerging are highlighted and labeled.

Table 2: Emerging clusters, with the paper closest to each cluster center as a human-readable example.

Cluster	Top terms	Papers	Growth	Example paper
COVID-19 and pandemic genomics	cov, sars, covid	213	5.0x	The COVID-19 pandemic...
Single-cell sequencing analysis	cell, single, seq	804	2.2x	Image-Based Single Cell Sorting Automation in Dr...
AI in clinical care	ai, health, healthcare	313	2.2x	Digital future in perioperative medicine: Are we...
Tumor immunology	tumor, cancer, immune	326	1.9x	A Sampling of Highlights from the Literature...
Genetic disease risk prediction	risk, disease, patients	404	1.6x	Varicose Veins Reach New Heights...
Cancer prognosis models	patients, survival, cancer	419	1.6x	Establishment of a 12-gene expression signature...
Variant calling and clinical sequencing	variants, sequencing, genome	481	1.5x	Computational methods of variant calling and the...
Drug target prediction	drug, discovery, prediction	713	1.4x	Drug discovery and drug marketing with the criti...

The fastest-growing front, COVID-19 genomics, is a reminder that research priorities move with events: machine learning was applied to viral genome surveillance, variant tracking, and drug repurposing within months. Pathogen genomics is also the front with the most direct biosecurity relevance, since the same tools that track variants can inform work on transmissibility. Single-cell analysis and clinical AI grew steadily and are larger, which makes them durable trends rather than spikes.

The clusters that lost share are informative too. *Microarray gene classification* and *feature selection for gene data* reflect an older generation of methods giving way to sequencing and deep learning. Protein sequence and structure prediction also lost share before 2020, just before AlphaFold2 changed the field, so extending the data to 2025 should show a sharp reversal there. One cluster, *gene expression programming*, is an evolutionary computing method that matched the query only because its name contains “gene expression.” It is a keyword false positive, and the clustering step surfaced it (Section 4).

3.3 Where the work is done

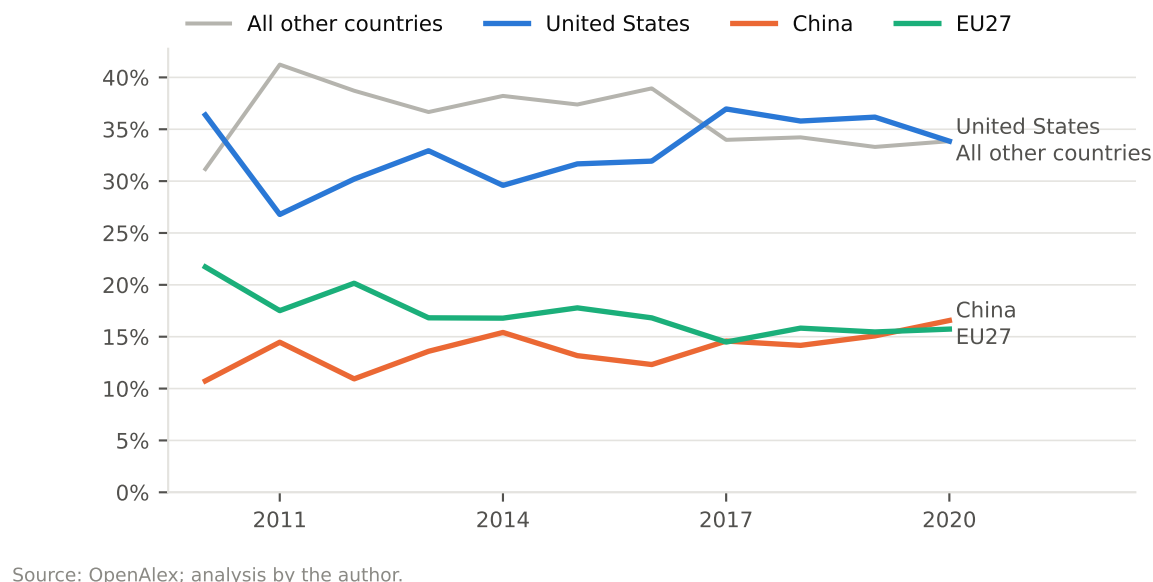


Figure 4: Share of world AI and biotechnology output by country bloc (fractional counting, sampling weighted).

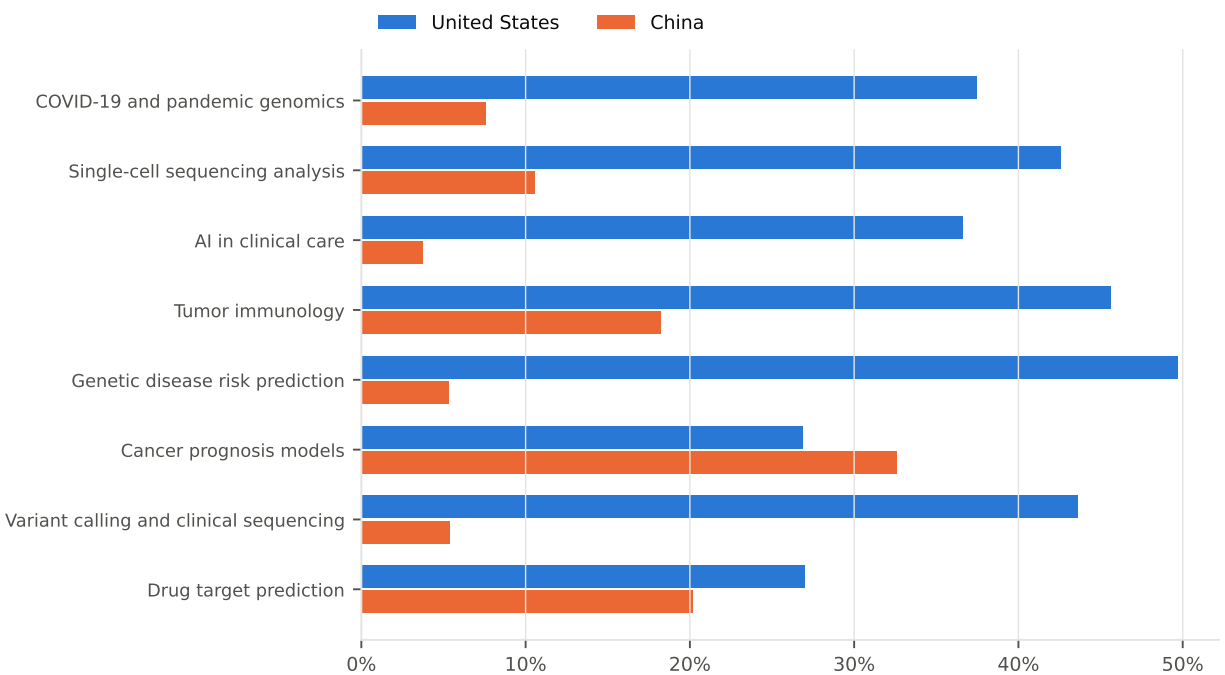
Table 3: Output and impact by bloc. The impact ratio compares a bloc’s share of top 10 percent papers with its share of all papers; values above 1 indicate disproportionately influential work.

Bloc	Share of papers	Share of top 10%	Share of top 1%	Impact ratio
United States	34.2%	35.1%	46.2%	1.02
Rest of world	20.4%	20.4%	16.2%	1.00
EU27	16.4%	17.4%	14.8%	1.06
China	14.0%	13.9%	11.9%	1.00

Table 3: Output and impact by bloc. The impact ratio compares a bloc’s share of top 10 percent papers with its share of all papers; values above 1 indicate disproportionately influential work.

Bloc	Share of papers	Share of top 10%	Share of top 1%	Impact ratio
United Kingdom	5.8%	6.6%	7.5%	1.14
India	5.0%	2.6%	0.7%	0.52
Japan	2.2%	1.9%	1.7%	0.89
South Korea	1.9%	2.1%	1.0%	1.08

3.3.1 Who leads the emerging fronts?

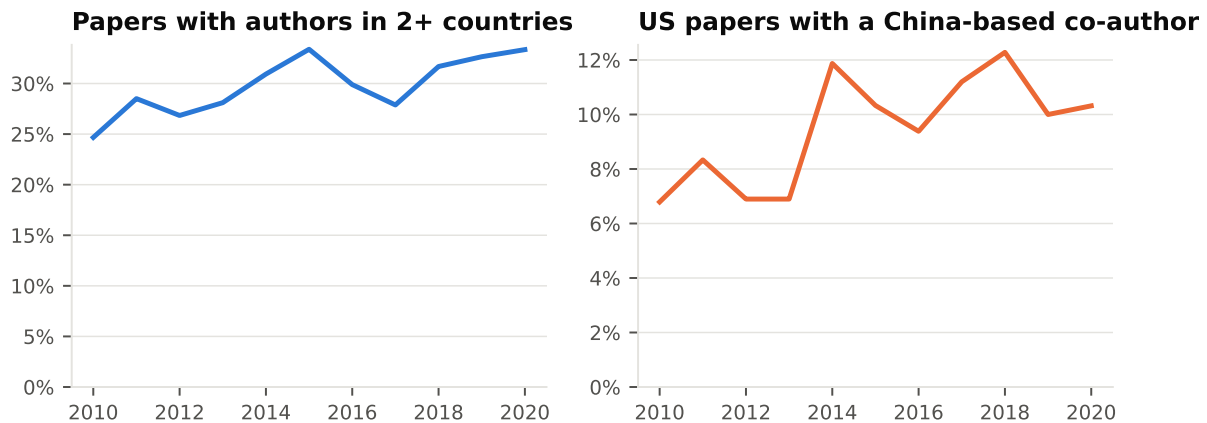


Source: OpenAlex; analysis by the author.

Figure 5: Share of output from the United States and China in each emerging cluster.

US institutions produce the larger share in 7 of the 8 emerging fronts, most clearly in genetic disease risk prediction and clinical sequencing. China leads in cancer prognosis models, and its share is highest in cancer-focused fronts. Since China’s overall share was still rising in 2020, this is the chart most likely to change when the analysis is extended to 2025.

3.4 Collaboration

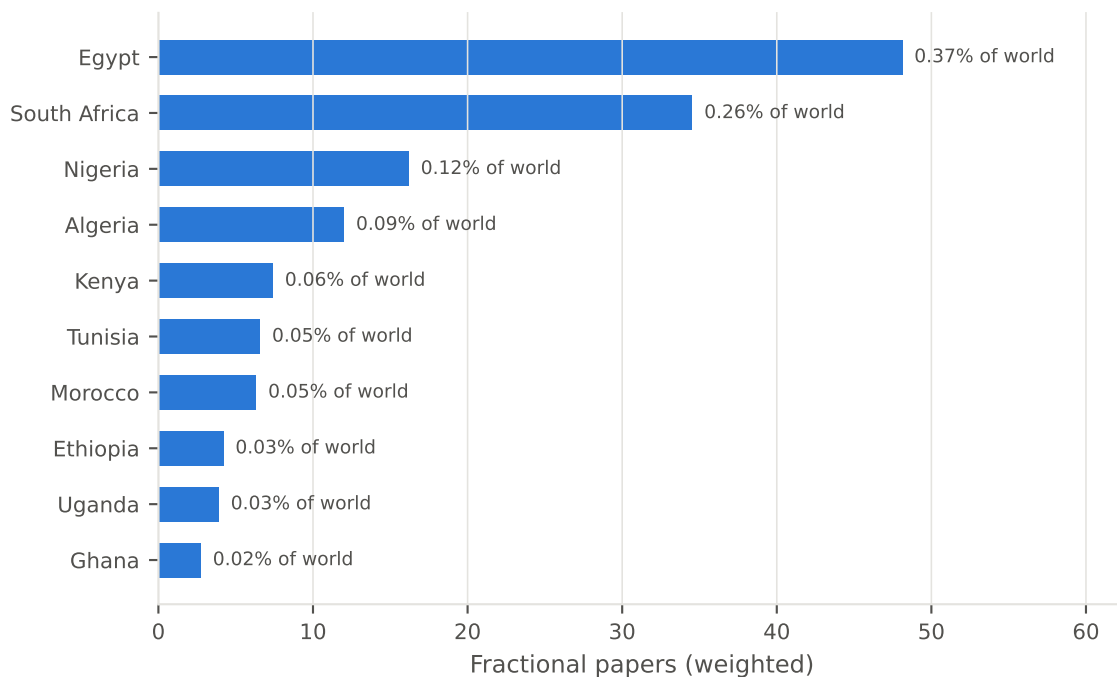


Source: OpenAlex; analysis by the author.

Figure 6: International collaboration (left) and the share of US papers that include a co-author based in China (right).

The share of papers with authors in more than one country rose from about a quarter to a third over the decade. US and China co-authorship rose through 2018 and then eased slightly in 2019 and 2020, the period when US research security scrutiny increased and the pandemic disrupted travel. These data show timing only; they cannot separate those causes, and the rise of Chinese output over the same years means the absolute number of joint papers still grew.

3.5 Global participation: Africa



Source: OpenAlex; analysis by the author.

Figure 7: Top African countries by fractional output, full period.

African institutions account for 1.5% of world output in 2020, led by Egypt, South Africa, and Nigeria, and 68% of African-authored papers that year include a partner outside the continent. The continent holds the world’s greatest human genetic diversity, yet contributes little of the AI research built on genomic data. Genomic AI models trained mostly on data from European-ancestry populations can perform worse for others, so where this research is done, and with whose data, is a question of both equity and scientific quality.

4 Data quality and validation

4.1 Affiliation entity resolution

Table 4: Entity resolution accuracy on a holdout of affiliation strings with a known institution, hidden from the matcher before matching.

Method	Predicted	Correct	Precision	Recall
All methods	1,354	1,230	90.8%	61.7%
Exact alias match	0	0	n/a	0.0%
Fuzzy name match	1,354	1,230	90.8%	61.7%

Table 5: Outcome for affiliation strings that OpenAlex left unresolved.

Outcome	Strings	Share of unresolved
Still unresolved	2,537	76.5%
Exact alias match	595	17.9%
Fuzzy name match	186	5.6%

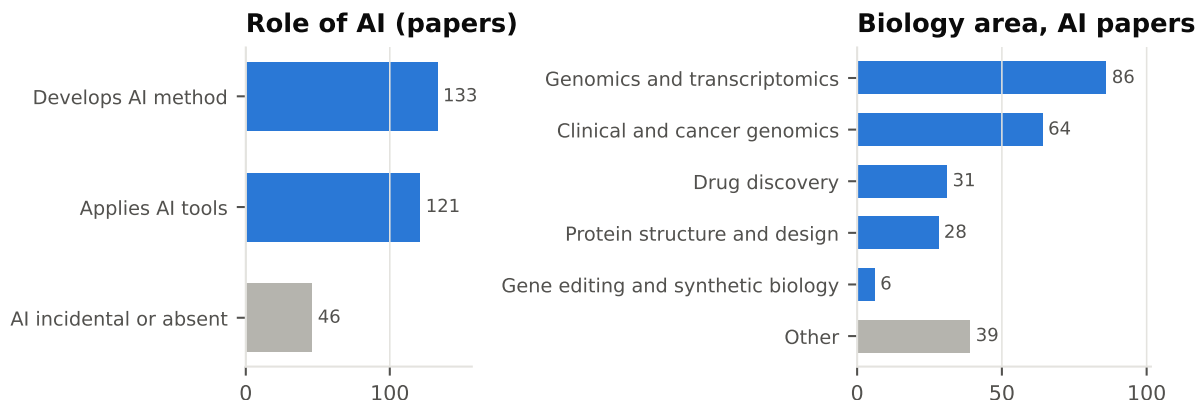
On the holdout, the fuzzy matcher was right 91% of the time when it made a match, and it matched 62% of strings. That trade-off is deliberate: a wrong institution is worse than a missing one, so the threshold favors precision. Applied to strings OpenAlex left unlinked, it recovered 24% of them. Recovered links add 1.0% to total country credit. Without this step, those authors would silently drop out of every country statistic above, and the loss is not evenly spread. Only 42% of the recovered credit belongs to North America and Europe, compared with 62% of the credit OpenAlex links itself, so unlinked affiliations come disproportionately from the rest of the world. Small as the total is, skipping this step would bias country comparisons against exactly the regions that are hardest to measure.

4.2 LLM-assisted annotation

A random sample of 300 papers (seeded, so it can be redrawn exactly) was annotated by Claude from each title and abstract, using the label scheme in `src/llm_classify.py`: does the paper *develop* an AI method, *apply* existing AI tools, or is AI incidental or absent? Each paper also received a biology area and a flag for pathogen or other dual-use biology, and records outside biology entirely were marked off topic.

Table 6: Corpus quality estimated from the annotated sample. Intervals are 95 percent Wilson confidence intervals.

Measure	Estimate	95% CI	n
Share of papers that are substantively about AI	84.7%	80.2% to 88.3%	300
Share about AI and about biology	83.0%	78.3% to 86.8%	300
Share involving pathogens or dual-use biology	3.3%	1.8% to 6.0%	300



Source: OpenAlex; annotation by Claude; analysis by the author.

Figure 8: Annotated sample: role of AI in each paper (left) and biology area among papers that use AI (right).

What it shows. About 85% of keyword matches are substantively about AI, and 83% are about both AI and biology. The 46 “incidental” papers are mostly reviews, commentaries, and wet-lab studies that mention machine learning once, or use “neural network” to mean the brain. 5 more are AI papers with no biology at all (steel alloy design, materials informatics, railway maintenance) that matched through phrases such as “materials genome” or “gene expression programming”, confirming what the clustering suggested. 3 records had a broken abstract (a journal table of contents, or an abstract from a different paper). Among AI papers, method development (44% of the sample) and application are roughly balanced. 10 papers (3.3%) involve pathogens, mostly viral genome classification and bacterial virulence or strain typing.

What it means for the numbers above. About 17% of the papers behind the yearly counts in Section 3.1 are not AI biology research, so levels are overstated. If that error rate is stable over time, growth rates and country shares are largely unaffected, but that should be checked year by year before it is assumed.

Validation still to do. An LLM’s labels are a measurement, not ground truth. The pipeline writes a template of 150 sampled papers (`data/labels/gold_labels.csv`) for hand labeling by a domain expert; rerunning `python src/llm_classify.py run` then reports agreement, Cohen’s kappa, and per-class precision and recall in this table. Until that is done, the estimates above should be read as Claude’s judgment, not as validated rates. These labels were produced by Claude within the analysis session; the scripted version sends the same prompt through the Anthropic API so the step can be rerun at scale.

4.3 Language coverage

Table 7: Publication language of the corpus.

Language	Share
en	98.3%
unknown	1.1%

Table 7: Publication language of the corpus.

Language	Share
fr	0.1%
id	0.1%
de	0.1%
es	0.1%

Nearly all records are in English. OpenAlex is English dominant, and the search terms are English, so research published only in Chinese, Japanese, Korean, or other languages is undercounted. A natural extension is to add translated search terms and non-English sources, which matters most for estimates of Chinese output.

5 Limitations

- **Partial paper-level coverage.** Country, cluster, and collaboration results cover 2010 to 2020, about 23% of all papers that match the query; 5 more years of records are needed to see the post-2020 surge at paper level.
- **Keyword definition.** The corpus depends on the term lists. About 17% of matches are not substantively about AI or biology (Section 4.2), and false negatives (AI papers that avoid these terms) are not measured.
- **Annotation not yet validated.** LLM labels have not yet been compared with hand labels.
- **Coverage.** OpenAlex coverage of abstracts, affiliations, and non-English venues is incomplete and has improved over time, which can create artificial trends.
- **Citation lag.** Recent papers have had less time to be cited; the latest year is excluded from impact measures and normalized percentiles are used.
- **Topic model.** Clusters use TF-IDF and latent semantic analysis. Neural sentence embeddings (supported in the code) usually give cleaner clusters.
- **Text clusters are not citation clusters.** Clusters reflect shared vocabulary. Citation-based clustering, as in CSET’s Map of Science, would capture intellectual lineage more directly.
- **Fractional counts reflect affiliation, not nationality** of researchers.

6 Reproducibility and scaling

All code is in a public GitHub repository: https://github.com/Batsi-2015/cset_sample. One command (`python src/run_pipeline.py`) rebuilds every table, and `quarto render` rebuilds this report. The pipeline is written so each step maps cleanly onto a production stack:

Local version	Production equivalent
<code>run_pipeline.py</code>	Airflow DAG (<code>dags/cset_sample_dag.py</code>) on Cloud Composer
Raw JSONL files	Google Cloud Storage bucket
DuckDB	BigQuery (views in <code>sql/</code> port with minor dialect changes)
Quality checks task	Airflow task that fails the run on bad data

Appendix: selected SQL

sql/views.sql

```
-- Reusable analytic views. Executed once after the pipeline builds the base
-- ↪ tables.

-- Country credit per paper, fractional counting.
-- Each author carries 1 / n_authors of the paper; an author with several
-- ↪ countries splits
-- that credit evenly. Credit is multiplied by the year's sampling weight so that
-- totals estimate the full OpenAlex corpus, not just the downloaded sample.
CREATE OR REPLACE VIEW v_country_credit AS
SELECT
  c.work_id,
  w.year,
  c.country_code,
  CASE
    WHEN c.country_code = 'US' THEN 'United States'
    WHEN c.country_code = 'CN' THEN 'China'
    WHEN r.eu27 = 'EU27' THEN 'EU27'
    WHEN c.country_code = 'GB' THEN 'United Kingdom'
    WHEN c.country_code = 'IN' THEN 'India'
    WHEN c.country_code = 'JP' THEN 'Japan'
    WHEN c.country_code = 'KR' THEN 'South Korea'
    ELSE 'Rest of world'
  END AS bloc,
  r.region,
  SUM(c.weight) AS credit,
  SUM(c.weight * w.sampling_weight) AS credit_weighted,
  w.top10pct,
  w.top1pct,
  w.citation_percentile
FROM author_countries_enriched c
JOIN works w USING (work_id)
LEFT JOIN country_regions r USING (country_code)
GROUP BY ALL;

-- Paper-level collaboration flags.
CREATE OR REPLACE VIEW v_work_collab AS
SELECT
  work_id,
  year,
  ANY_VALUE(sampling_weight) AS
-- ↪ sampling_weight,
  COUNT(DISTINCT country_code) AS
-- ↪ n_countries,
  BOOL_OR(country_code = 'US') AS has_us,
```

```

        BOOL_OR(country_code = 'CN') AS has_cn,
        BOOL_OR(region = 'Africa') AS
    ↪ has_africa,
        BOOL_OR(region IS DISTINCT FROM 'Africa') AS
    ↪ has_non_africa
FROM (
    SELECT c.work_id, w.year, w.sampling_weight, c.country_code, r.region
    FROM author_countries_enriched c
    JOIN works w USING (work_id)
    LEFT JOIN country_regions r USING (country_code)
)
GROUP BY work_id, year;

```

sql/country_impact.sql

```

-- Output versus impact by bloc.
-- impact_ratio > 1 means a bloc's share of the top 10% most cited papers (field
    ↪ and year
-- normalized by OpenAlex) is larger than its share of all papers.
-- The most recent year is excluded because citations have not had time to
    ↪ accumulate.
WITH base AS (
    SELECT bloc,
           SUM(credit_weighted) AS output,
           SUM(CASE WHEN top10pct THEN credit_weighted ELSE 0 END) AS
    ↪ output_top10,
           SUM(CASE WHEN top1pct THEN credit_weighted ELSE 0 END) AS
    ↪ output_top1
    FROM v_country_credit
    WHERE year BETWEEN {impact_start} AND {impact_end}
    GROUP BY bloc
)
SELECT
    bloc,
    output,
    output / SUM(output) OVER () AS
    ↪ share_all,
    output_top10 / NULLIF(SUM(output_top10) OVER (), 0) AS
    ↪ share_top10,
    output_top1 / NULLIF(SUM(output_top1) OVER (), 0) AS
    ↪ share_top1,
    (output_top10 / NULLIF(SUM(output_top10) OVER (), 0))
    / NULLIF(output / SUM(output) OVER (), 0) AS
    ↪ impact_ratio
FROM base
ORDER BY output DESC;

```